

ME EN 7220 - System ID

Homework 4 , Open Loop Calibration

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Abstract—Parameters for a serial link manipulator defined by a blend of Denavit-Hartenberg and Hayati parameters are calibrated using noisy joint and effector pose data using open loop calibration and iterative least squares.

I. INTRODUCTION

Open loop calibration is a means for estimating the parameters of a manipulator by fitting noisy input and output data to a forward kinematics model of a serial link manipulator.

II. RANDOM POSES

A. Simulation

Rewrite the Matlab program `random_poses.m` for an arbitrary n -DOF manipulator, that simulates full pose measurement in open-loop calibration. Coordinate systems are set up to locate the metrology system and the measured end link frame. The parameters are mixed between DH parameters and Hayati parameters.:

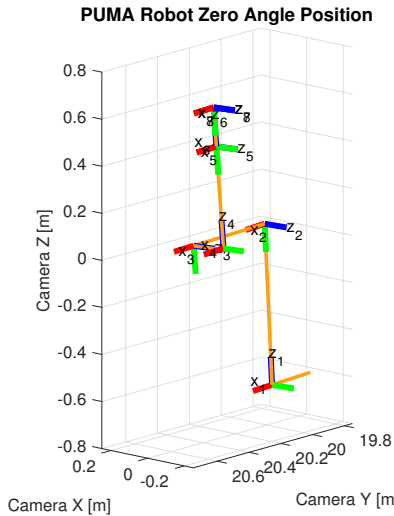


Fig. 1: Zero-Angle Position for the specified PUMA robot

`random_poses.m` is changed to deal with a mixture of Denavit-Hartenberg and Hayati, with either parameter set assigned to each joint. Two corners of an N -dimensional joint

space are defined by the user, and random joint measurements are generated between these upper and lower bounds. The parameters for the robot model are defined by ϕ , the rows of which can be specify either Denavit-Hartenberg and Hayati parameters. Using the randomized measurements generated above and robot model parameter ϕ , `FK_blend.m` performs the forward kinematics for the robot (including camera and tool frame “joints”). The per-joint transformations performed are shown in Equations 1 to ???. `FK_blend.m` returns the effector position, orientation, and pose as a homogeneous transformation.

The robot that will be simulated / calibrated is a 6 DOF serial manipulator modeled after PUMA designs. Its parameters are given in Table I.

TABLE I: Blended DH/Hayati Specification for the PUMA Robot

j	γ_j	α_i	a_j	d_j	β_j
0	$\pi/2$	$\pi/100$	20	0	$\pi/100$
1	0	$-\pi/2$	0	0.6858	10
2	0	$\pi/100$	0.4318	0	$\pi/100$
3	0	$\pi/2$	-0.0203	0.1491	10
4	0	$-\pi/2$	0	0.4331	10
5	0	$\pi/2$	0	0	10
6	0	$-\pi/2$	0	0.1655	10
7	0	0	0	0	10

D-H Parameter Transformations

$$\mathbf{v}_{disp} = \mathbf{R}_z(\theta_j) \begin{bmatrix} a_j \\ 0 \\ 0 \end{bmatrix} \quad (1)$$

$$\mathbf{R}_j = \mathbf{R}_z(\theta_j) \mathbf{R}_x(\alpha_j) \quad (2)$$

$$\mathbf{T}_j = \begin{bmatrix} \mathbf{R}_j & \mathbf{v}_{disp} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

Hayati Transformations

$$\mathbf{v}_{disp} = \mathbf{R}_z(\theta_j) \begin{bmatrix} a_j \\ 0 \\ 0 \end{bmatrix} \quad (4)$$

$$\mathbf{R}_j = \mathbf{R}_z(\theta_j) \mathbf{R}_x(\alpha_j) \mathbf{R}_y(\beta_j) \quad (5)$$

$$\mathbf{T}_j = \begin{bmatrix} & \mathbf{R}_j & \mathbf{v}_{disp} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

Each $\mathbf{T}_j$ is applied in turn, according to whether Hayati or DH parameters are to be used. Use of Hayati parameters for a serial link is indicated by the condition $-\pi \leq \beta_j \leq \pi$, in which case d_j is ignored. When β_j is outside this range, the parameters are interpreted as DH and β_j is ignored. Noise is then added to both the position and orientation measurements in the same manner as the `random_poses.m` function presented by Hollerbach in [1]. Randomly generated configurations and pose measurements can be seen in Figures 2 and 3, respectively.

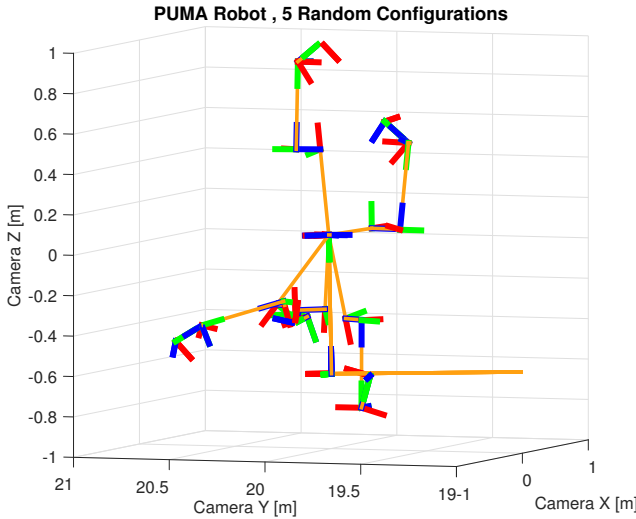


Fig. 2: A selection of random robot poses, without noise

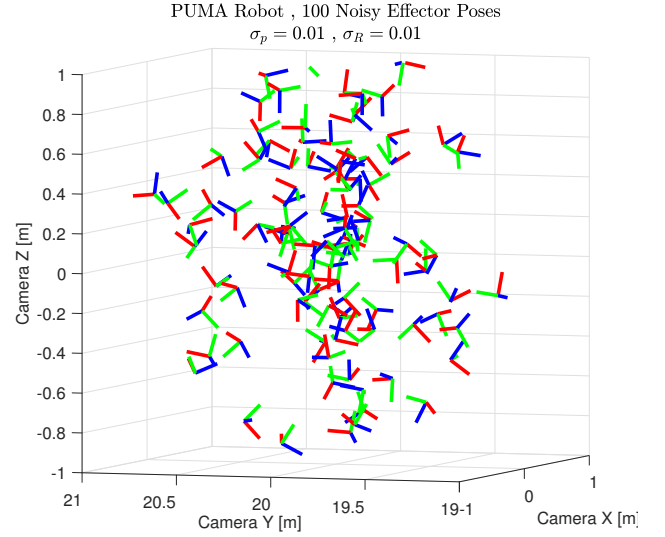


Fig. 3: A selection of 100 random simulated robot pose measures, with noise

III. OPEN LOOP CALIBRATION

A. DH Pose Error

As shown in [1], the pose error for a serial manipulator defined with DH parameters is shown in equation 7.

$$\begin{bmatrix} \Delta \mathbf{d}^i \\ \Delta \mathbf{r}^i \end{bmatrix} = \sum_{j=0}^{n+1} \Delta \gamma_j \begin{bmatrix} \Delta \mathbf{z}_{j-1}^i \times \mathbf{d}_{j-1,n+1} \\ \Delta \mathbf{z}_{j-1}^i \end{bmatrix} + \sum_{j=0}^n \Delta \alpha_j \begin{bmatrix} \Delta \mathbf{x}_j^i \times \mathbf{d}_{j,n+1} \\ \Delta \mathbf{x}_j^i \end{bmatrix} + \sum_{j=0}^n \Delta a_j \begin{bmatrix} \mathbf{x}_j^i \\ \mathbf{0} \end{bmatrix} + \sum_{j=0}^{n+1} \Delta d_j \begin{bmatrix} \mathbf{z}_{j-1}^i \\ \mathbf{0} \end{bmatrix} \quad (7)$$

The Parameter Jacobian for DH Parameter estimation is shown in Equation 5.31 of [1]. Note that in the above and throughout this work, the angle measurement gain k_j is not calibrated. It is assumed that angle measurements ψ_i differ from the actual joint angle only by an offset γ_i .

B. Hayati Parameter Jacobian

Similar to the above, we can formulate the total rotation error for a manipulator defined by Hayati parameters with Equation 8 by considering each of the parameters to be a joint, each with a position and orientation contribution to pose estimation error.

$$\Delta \mathbf{r}^i = \sum_{j=0}^n \mathbf{x}_j'' \Delta \alpha_j + \sum_{j=0}^n \mathbf{y}_j' \Delta \beta_j + \sum_{j=0}^{n+1} \mathbf{z}_{j-1} \Delta \gamma_j \quad (8)$$

The total position error is formed in the same way, with $\mathbf{x}_j'' \Delta a_j$ as the error along $\mathbf{x}''$, $\mathbf{x}_j'' \Delta \alpha_j \times \mathbf{d}_{j,n+1}$ as the

error about $\mathbf{x}''$, $\mathbf{y}'_j \Delta \beta_j \times \mathbf{d}_{j,n+1}$ as the error about $\mathbf{y}'_j$, and $\mathbf{z}_{j-1} \Delta \gamma_j \times \mathbf{d}_{j-1,n+1}$ as the error about $\mathbf{z}_{j-1}$.

$$\begin{aligned} \Delta \mathbf{d}^i = & \sum_{j=0}^n [\mathbf{x}''_j \Delta a_j + \mathbf{x}''_j \Delta \alpha_j \times \mathbf{d}_{j,n+1}] \\ & + \sum_{j=0}^n [\mathbf{y}'_j \Delta \beta_j \times \mathbf{d}_{j,n+1}] \\ & + \sum_{j=0}^{n+1} [\mathbf{z}_{j-1} \Delta \gamma_j \times \mathbf{d}_{j-1,n+1}] \end{aligned} \quad (9)$$

Equations 8 and 9 can be combined into a single expression of pose error, Equation 10.

$$\begin{aligned} \begin{bmatrix} \Delta \mathbf{d}^i \\ \Delta \mathbf{r}^i \end{bmatrix} = & \sum_{j=0}^{n+1} \Delta \gamma_j \begin{bmatrix} \mathbf{z}_{j-1} \times \mathbf{d}_{j-1,n+1} \\ \mathbf{z}_{j-1} \end{bmatrix} \\ & + \sum_{j=0}^n \Delta \alpha_j \begin{bmatrix} \mathbf{x}''_j \times \mathbf{d}_{j,n+1} \\ \mathbf{x}''_j \end{bmatrix} \\ & + \sum_{j=0}^n \Delta a_j \begin{bmatrix} \mathbf{x}''_j \\ \mathbf{0} \end{bmatrix} \\ & + \sum_{j=0}^n \Delta \beta_j \begin{bmatrix} \mathbf{y}'_j \times \mathbf{d}_{j,n+1} \\ \mathbf{y}'_j \end{bmatrix} \end{aligned} \quad (10)$$

with

$$\mathbf{x}''_j = \mathbf{R}_z(\theta_j) \mathbf{x}_{j-1} \quad (11)$$

$$\mathbf{y}'_j = \mathbf{R}_z(\theta_j) \mathbf{R}_x(\alpha_j) \mathbf{x}_{j-1} \quad (12)$$

$$\mathbf{z}_j = \mathbf{R}_z(\theta_j) \mathbf{R}_x(\alpha_j) \mathbf{R}_y(\beta_j) \mathbf{z}_{j-1} \quad (13)$$

The Jacobian can be formed

$$\Delta \mathbf{y}^i = \mathbf{J}_v^i \Delta \phi = \begin{bmatrix} \mathbf{J}_{v,\gamma}^i & \mathbf{J}_{v,\alpha}^i & \mathbf{J}_{v,a}^i & \mathbf{J}_{v,\beta}^i \end{bmatrix} \phi \quad (14)$$

$$\mathbf{J}_{v,\gamma}^i = \begin{bmatrix} \mathbf{z}_{-1} \times \mathbf{d}_{-1,n+1} & \cdots & \mathbf{z}_n \times \mathbf{d}_{n,n+1} \\ \mathbf{z}_{-1} & \cdots & \mathbf{z}_n \end{bmatrix} \quad (15)$$

$$\mathbf{J}_{v,\alpha}^i = \begin{bmatrix} \mathbf{x}''_0 \times \mathbf{d}_{0,n+1} & \cdots & \mathbf{x}''_n \times \mathbf{d}_{n,n+1} \\ \mathbf{x}''_0 & \cdots & \mathbf{x}''_n \end{bmatrix} \quad (16)$$

$$\mathbf{J}_{v,a}^i = \begin{bmatrix} \mathbf{x}''_0 & \cdots & \mathbf{x}''_n \\ \mathbf{0} & \cdots & \mathbf{0} \end{bmatrix} \quad (17)$$

$$\mathbf{J}_{v,\beta}^i = \begin{bmatrix} \mathbf{y}'_0 \times \mathbf{d}_{0,n+1} & \cdots & \mathbf{y}'_n \times \mathbf{d}_{n,n+1} \\ \mathbf{y}'_0 & \cdots & \mathbf{y}'_n \end{bmatrix} \quad (18)$$

C. Blended DH / Hayati Calibration

$$\Delta \mathbf{y}^i = \mathbf{J}_v^i \Delta \phi = \begin{bmatrix} \mathbf{J}_{v,\gamma}^i & \mathbf{J}_{v,\alpha}^i & \mathbf{J}_{v,a}^i & \mathbf{J}_{v,d}^i & \mathbf{J}_{v,\beta}^i \end{bmatrix} \phi \quad (19)$$

A parameter Jacobian suitable for both the DH and the Hayati follows straightforwardly from the above. The following

rules apply to the construction of the Combined Parameter Jacobian, Equation 19.

DH Case:

- Interpret $\mathbf{J}_{v,\alpha}^i$ as presented in [1].
- $\mathbf{J}_{v,\beta}^i = [0 \ 0 \ 0 \ 0 \ 0 \ 0]^T$

Hayati Case:

- Interpret $\mathbf{J}_{v,\alpha}^i$ as shown in Equation 18
- $\mathbf{J}_{v,d}^i = [0 \ 0 \ 0 \ 0 \ 0 \ 0]^T$

Gauss-Newton Nonlinear Least Squares can be used to fit the combined DH / Hayati parameter model to measurements of the effector position and orientation. The regressor matrix $\mathbf{C}$ is formed by stacking the parameter Jacobians for each measurement, Equation 21. An initial parameter guess ϕ^1 is needed for the first iteration. This was guess was obtained by perturbing the model in Table I with the `perturb_dh_hayati` MATLAB function. Ordinary Least Squares is used to obtain $\Delta \phi$, the adjustment made to the parameter estimate ϕ that minimizes error about the current linearization of $\mathbf{f}$. (Equation 22)

The next iteration proceeds at the adjusted ϕ^k . $\mathbf{p}^i$ is the difference between the measured marker position and the position of the marker predicted by forward kinematics (Eq. 3 & 6) as a function of the joint measurements ψ^i and the current estimate of the parameters ϕ^k . $\mathbf{R}^i$ is the difference between the measured marker orientation and the position of the effector predicted by forward kinematics as a function of the joint measurements ψ^i and the current estimate of the parameters ϕ^k . The difference in orientation is given by the Differential Orthogonal Rotation (dor), presented by Hollerbach in [1]. It takes as input two rotation matrices, and provides as output the three magnitudes of principal rotations, assumed to be transitive. It is applicable to our problem as long as the small angle assumption is valid. Iteration proceeds until $\|\Delta \phi\|$ falls below some threshold value, set to 0.00001 in this case.

$$\mathbf{f}^i(\phi^k + \Delta \phi) - \mathbf{f}^i(\phi^k) = \frac{\partial \mathbf{f}^i}{\partial \phi} \Delta \phi = \mathbf{J}^i \Delta \phi \quad (20)$$

$$\mathbf{C} = \begin{bmatrix} \vdots \\ \mathbf{J}^i \\ \vdots \end{bmatrix} \quad (21)$$

$$\Delta \phi = (\mathbf{C}^T \mathbf{C})^{-1} \mathbf{C}^T \begin{bmatrix} \cdots & \Delta \mathbf{p}^i & \Delta \mathbf{R}^i & \cdots \end{bmatrix} \quad (22)$$

$$\Delta \mathbf{p}^i = \mathbf{p}_{\text{measured}} - \mathbf{p}_{\text{FK}} \quad (23)$$

$$\Delta \mathbf{R}^i = \text{dor}(\mathbf{R}_{\text{measured}}, \mathbf{R}_{\text{FK}}) \quad (24)$$

$$\phi^{k+1} = \phi^k + \Delta \phi \quad (25)$$

D. Characterization of Error

Error reported for each fit is the median position error and median orientation error between the effector frame measurements and the effector pose predicted by the forward kinematics for the final parameter estimate.

Position error is defined to be the Euclidean distance between the noisy position measurement of the effector and the position of the effector predicted by forward kinematics as a function of the joint measurements ψ^i and the final estimate of the parameters ϕ , $\|P_m - P_c\|$.

Orientation error is defined to be the L2-Norm of $\Delta\epsilon$, which is defined in Equation 3.91 of [2], repeated here as Equation 26. $\eta(q)$ returns the scalar portion of quaternion q and $\epsilon(q)$ returns the vector portion: See Equations 27 and 28. Note that rotation matrices must be converted to quaternions to apply Equation 26.

$$\Delta\epsilon = \eta(q_e)\epsilon(q_d) - \eta(q_d)\epsilon(q_e) - S(\epsilon(q_d))\epsilon(q_e) \quad (26)$$

$$\eta(q) = q_0 \quad (27)$$

$$\epsilon(q) = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} \quad (28)$$

$\|\Delta\epsilon\|$ takes the value of 0 when the two orientations are exactly aligned, and 1 when any of the basis vectors of the orientations are exactly opposed. These properties make $\|\Delta\epsilon\|$ an easily-understood scalar measure of the magnitude of difference between two orientations.

IV. RESULTS

A. PUMA Serial Robot

Write and test a Matlab program `fullfit.m` that solves for DH and Hayati parameters in open-loop calibration using full pose measurement. The parameters are mixed between DH parameters and Hayati parameters.:

Using the parameters from Table I and the above `random_poses.m` function, 600 noisy effector pose measurements were generated with $-\pi \leq \psi_j^i \leq \pi$, and with noise added to position and orientation, respectively $\sigma_p = 0.01$, $\sigma_R = 0.01$.

The full fit procedure described above is applied to the noisy data, with the resulting parameter estimates shown in Table II. The estimated parameters match the ideal parameters very closely. The initial guess was generated using `perturb_dh_hayati` with increasing values of σ_{perturb} . See Table III.

TABLE II: Estimated Parameters for the PUMA Robot

j	γ_j	α_i	a_j	d_j	β_j
0	1.5708	0.0312	20.0008	0.1357	0.0313
1	0.0001	-1.5709	0.0006	0.6830	9.7780
2	0.0005	0.0314	0.4321	-0.0109	0.0311
3	-0.0001	1.5708	-0.0207	0.1489	9.9581
4	0.0002	-1.5710	0.0002	0.4329	10.1942
5	-0.0001	1.5711	-0.0000	0.0005	9.8658
6	-0.0007	-1.5710	-0.0007	0.1653	10.0478
7	-0.0002	0	0	-0.0003	10.0000

It was noted that some of the $\beta_j > \pi$ “flag” parameters drifted slightly during calibration, though they have no impact on forward kinematics. None of these drifted into the $-\pi \leq \beta_j \leq \pi$ range during iteration. Iteration limit was set to 100.

TABLE III: Parameter Perturbation and Pose Error

σ_{perturb}	$\ P_m - P_c\ $	$\ \Delta\epsilon\ $	Iterations
0.0250	0.0158	0.0033	5
0.0500	0.0158	0.0033	5
0.0500	0.0158	0.0033	5
0.1000	0.0158	0.0033	6
0.2500	0.0158	0.0033	7
0.5000	0.0158	0.0033	9
0.7500	0.4049	0.6505	LIMIT
1.0000	0.4502	0.6678	LIMIT
2.0000	0.4051	0.6473	LIMIT

Calibration with initial guesses $\sigma_{\text{perturb}} > 0.5$ did not converge.

V. CONCLUSION

The estimated parameters provide calculated effector poses reasonably close to those measured. The method described allow us to select the appropriate geometry representation for each link individually, and also to calibrate all links together with the same procedure with estimates that closely match the actual.

REFERENCES

- [1] J. M. Hollerbach, “Open loop kinematic calibration,” in *System Identification for Robotics*, ch. 5, Salt Lake City: University of Utah, 2018.
- [2] B. Siciliano, L. Sciavicco, L. Villani, and G. Oriolo, “Robotics—modelling, planning and control. advanced textbooks in control and signal processing series,” 2009.